FATIMA JINNAH WOMEN UNIVERSITY

DEPARTMENT OF SOFTWARE ENGINEERING

Course: Cloud Computing (BSE-410)
Program: BS(SE)-V
Assignment No.: 01
Assignment Title: Git, Gitea, GitHub, Git LFS, and GitHub Pages
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Submission Date: 10 October 2026
Total Marks: 10

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1. Assignment Objective

The objective of this assignment is to install and run Gitea on an Ubuntu server, create and manage a Git repository, push the same repository to Gitea and GitHub, track three large files using Git Large File Storage (Git LFS), and publish a personal portfolio or CV using GitHub Pages.

By completing this assignment, I will learn how to:

- Install and run Gitea on an Ubuntu server.
- Create and manage Git repositories.
- Work with multiple Git remotes.
- Authenticate safely without exposing credentials.
- Track and push large files using Git LFS.
- Create and publish a portfolio website using GitHub Pages.

2. Required Environment

The following environments are required to complete this assignment:

- **Ubuntu Server:** Gitea installation and Git terminal operations for Tasks 1, 2, and 3.
- **Web Browser:** Accessing Gitea, GitHub, and GitHub Pages.

- **GitHub Repository:** Storing the assignment evidence in the existing public repository named `cc`.

The required Ubuntu terminal prompt is:

```
hifzashakeel52@ubuntu:~$
```

3. Getting Started

First, open the existing `cc` repository on the Ubuntu server and create the required assignment directory:

```
mkdir -p Assignments/Assignment01/screenshots
```

Ensure that Git and Git LFS are installed. Docker and Docker Compose will be installed during Task 1.

The assignment directory must contain:

- `Assignment01.md`
- `Assignment01_Solution.docx`
- `Assignment01_Solution.pdf`
- `screenshots/`

The screenshots folder must contain all 12 screenshots specified in this report.

4. Task 1: Install Gitea and Push a Repository from the Ubuntu Server

4.1 Install and Run Gitea

Step 1: Update Ubuntu packages and install Docker and curl.

```
sudo apt update -y
sudo apt install -y docker.io curl
```

Step 2: Install the Docker Compose plugin.

```
sudo mkdir -p /usr/local/lib/docker/cli-plugins
sudo curl -SL
https://github.com/docker/compose/releases/latest/download/docker-compose-
linux-x86_64 -o /usr/local/lib/docker/cli-plugins/docker-compose
sudo chmod +x /usr/local/lib/docker/cli-plugins/docker-compose
```

Step 3: Start Docker and configure permissions.

```
sudo systemctl start docker
sudo usermod -aG docker $USER
exit
```

Completely sign out of the Ubuntu server and reconnect so the updated group membership takes effect.

Step 4: Verify the installations.

```
git --version
docker --version
docker compose version
```

All three commands should display their installed versions.

Step 5: Download the instructor-provided Gitea setup repository.

```
cd ~
git clone https://github.com/WaqasSaleem97/Gitea.git
cd Gitea
```

Step 6: Start Gitea and PostgreSQL.

```
docker compose up -d
```

Step 7: Check the running containers.

```
docker compose ps
```

Both Gitea and PostgreSQL should be running.

Step 8: Check the Gitea HTTP response.

```
curl -sS -o /dev/null -w "Gitea HTTP status: %{http_code}\n"
http://127.0.0.1:3000/
```

Step 9: Find the Ubuntu server IP address.

```
ip addr
```

Open the following address in a browser, replacing the placeholder with the actual server IP:

```
http://<ubuntu-server-ip>:3000/
```

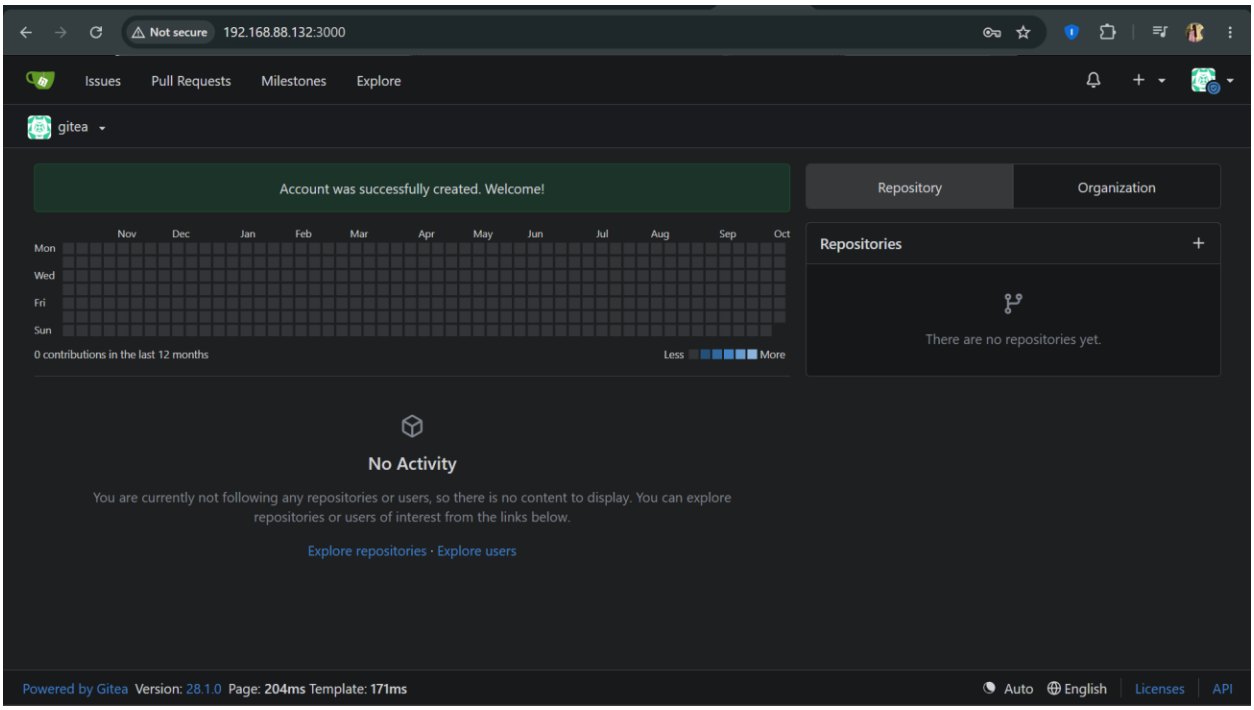
Complete the initial Gitea installation using these database settings:

Setting	Required Value
Database type	PostgreSQL

Database host	`db:5432`
Database username	`gitea`
Database password	`gitea`
Database name	`gitea`
Server domain	Ubuntu server IP
Gitea base URL	`http://<ubuntu-server-ip>:3000/`

Create the Gitea administrator account and finish the installation. Do not show passwords or tokens in screenshots.

Screenshot 1: Gitea Running



Filename: `task1_gitea_running.png`

The screenshot should show the Ubuntu prompt, `docker compose ps`, both containers running, and the successful Gitea HTTP status.

4.2 Create and Push the Initial Repository

Step 1: Create an empty public repository in Gitea named exactly `Assignment01`. Do not initialize it with a README, `.gitignore`, or license.

Step 2: Create a separate local repository on the Ubuntu server.

```
cd ~
mkdir Assignment01
cd Assignment01
git init
git branch -M main
```

Step 3: Create the README file.

```
printf "# Hifza Shakeel\n\nRegistration Number: fa24b1-se-052\n" > README.md
```

Step 4: Add and commit the README.

```
git add README.md
git commit -m "Add student information"
```

Step 5: Create a Gitea personal access token with repository read/write permission (`write:repository`). Use `ubuntu-server-token` as its name.

Step 6: Add the Gitea remote.

Replace `<gitea-username>` with your actual Gitea account username.

```
git remote add gitea http://<gitea-username>@127.0.0.1:3000/<gitea-
username>/Assignment01.git
```

Step 7: Verify the remote.

```
git remote -v
```

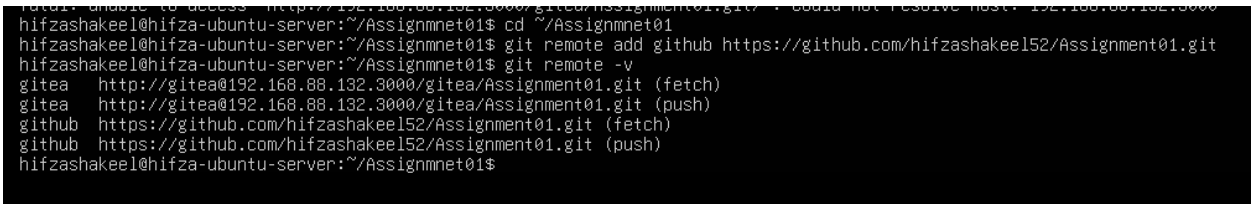
Step 8: Push the repository to Gitea.

```
git push -u gitea main
```

Enter the personal access token only when Git asks for the password. Never place it in the remote URL or command.

Screenshot 2: Push to Gitea

Filename: `task1_gitea_push.png`



The screenshot should show the remote, push command, successful result, and Ubuntu prompt. The token must not be visible.

4.3 Verify the Gitea Repository

Open the `Assignment01` repository in Gitea.

Verify that:

- The repository owner and name are visible.
- The repository is public.
- The README displays your full name and registration number.
- The Gitea URL is visible.

Screenshot 3: Gitea Repository

Filename: `task1_gitea_repository.png`



Task 1 Result: After successful completion, the repository will be stored in Gitea and accessible through the Ubuntu server's Gitea service.

5. Task 2: Push the Same Repository to GitHub

5.1 Create the GitHub Repository and Configure Remotes

Create an empty public GitHub repository named exactly `Assignment01`. Do not initialize it with a README, `.gitignore`, or license.

Open the same local repository on Ubuntu:

```
cd ~/Assignment01
```

Add GitHub as the second remote:

```
git remote add github https://github.com/hifzashakeel52/Assignment01.git
```

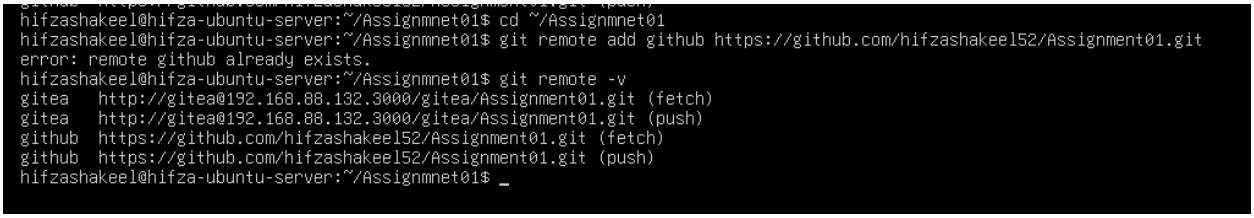
Verify both remotes:

```
git remote -v
```

The output should show both `gitea` and `github`, including their fetch and push URLs. Neither URL should contain a password or token.

Screenshot 4: Both Git Remotes

Filename: `task2_remotes.png`



5.2 Push the Repository to GitHub

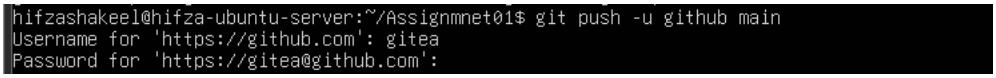
Run:

```
git push -u github main
```

Authenticate safely if prompted. Confirm that the push completes successfully.

Screenshot 5: GitHub Push

Filename: `task2_github_push.png`



5.3 Verify the GitHub Repository

Open:

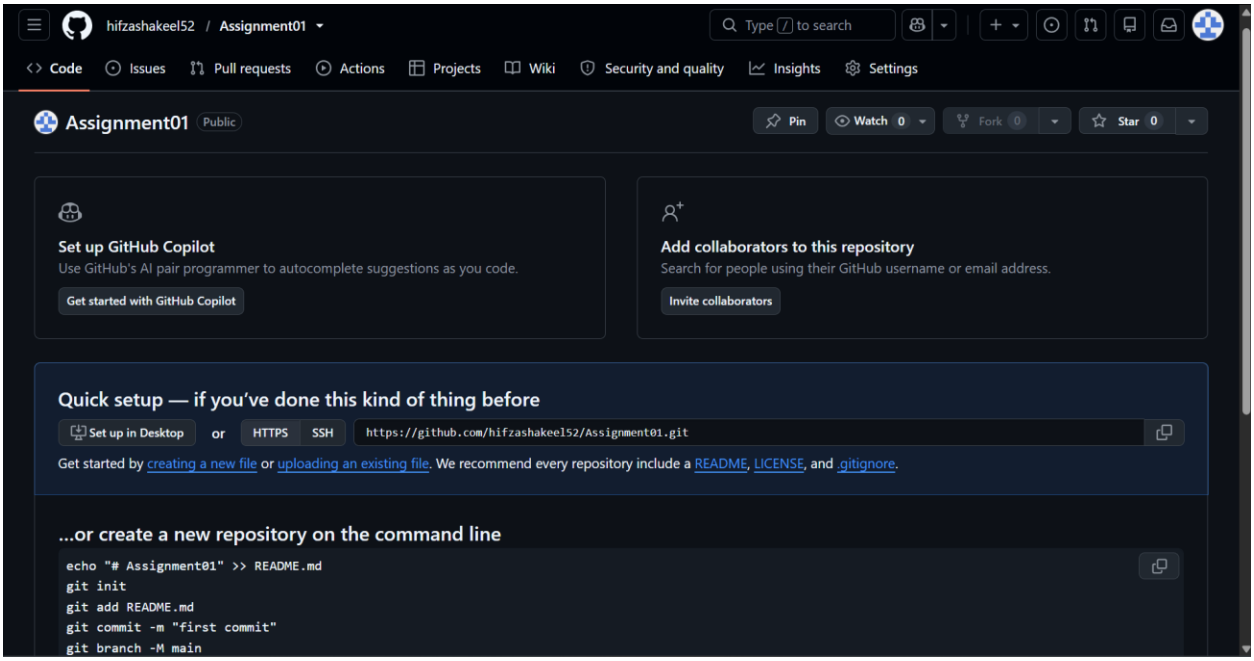
```
https://github.com/hifzashakeel52/Assignment01
```

Verify that:

- The repository owner is `hifzashakeel52`.
- The repository name is `Assignment01`.
- The repository is public.
- The rendered README displays your name and registration number.

Screenshot 6: GitHub Repository

Filename: `task2_github_repository.png`



Task 2 Result: After successful completion, the same local repository will be connected to Gitea and GitHub and available on GitHub.

6. Task 3: Track Three Large Files with Git LFS

6.1 Configure Git LFS

Open the local repository:

```
cd ~/Assignment01
```

Initialize Git LFS:

```
git lfs install
```

Check the Git LFS version:

```
git lfs version
```

Configure Git LFS to track `.bin` files:

```
git lfs track "*.bin"
```

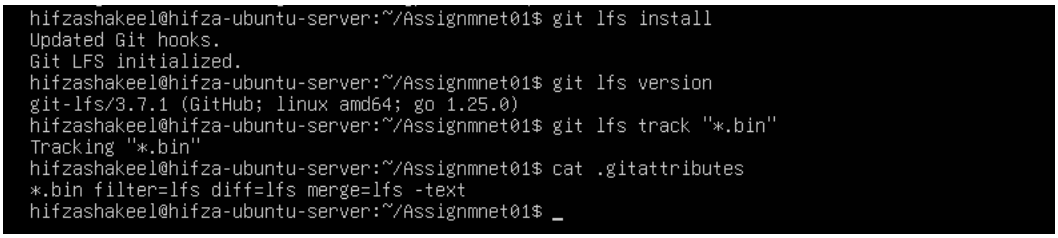
Display the tracking rule:

```
cat .gitattributes
```

The output should contain the `*.bin` tracking rule.

Screenshot 7: Git LFS Setup

Filename: `task3_lfs_setup.png`



6.2 Create and Verify Three Large Files

Create three files larger than 100 MiB each:

```
truncate -s 101M large-file-1.bin
truncate -s 102M large-file-2.bin
truncate -s 103M large-file-3.bin
```

Add the files and the Git LFS configuration:

```
git add .gitattributes large-file-1.bin large-file-2.bin large-file-3.bin
```

Verify the tracked files and their sizes:

```
git lfs ls-files --size
```

Confirm that all three files are listed and each is larger than 100 MiB.

Screenshot 8: Three LFS Files

Filename: `task3_lfs_files.png`

```
hifzashakeel@hifza-ubuntu-server:~/Assignmnet01$ git lfs version
git-lfs/3.7.1 (GitHub; linux amd64; go 1.25.0)
hifzashakeel@hifza-ubuntu-server:~/Assignmnet01$ git lfs track "*.bin"
Tracking "*.bin"
hifzashakeel@hifza-ubuntu-server:~/Assignmnet01$ cat .gitattributes
*.bin filter=lfs diff=lfs merge=lfs -text
hifzashakeel@hifza-ubuntu-server:~/Assignmnet01$ truncate -s 101M large-file-1.bin
hifzashakeel@hifza-ubuntu-server:~/Assignmnet01$ truncate -s 102M large-file-2.bin
hifzashakeel@hifza-ubuntu-server:~/Assignmnet01$ truncate -s 103M large-file-3.bin
hifzashakeel@hifza-ubuntu-server:~/Assignmnet01$ ls -lh *.bin
-rw-rw-r-- 1 hifzashakeel hifzashakeel 101M Oct 10 11:13 large-file-1.bin
-rw-rw-r-- 1 hifzashakeel hifzashakeel 102M Oct 10 11:13 large-file-2.bin
-rw-rw-r-- 1 hifzashakeel hifzashakeel 103M Oct 10 11:13 large-file-3.bin
hifzashakeel@hifza-ubuntu-server:~/Assignmnet01$ git add .gitattributes large-file-1.bin large-file-2.bin large-file-3.bin
hifzashakeel@hifza-ubuntu-server:~/Assignmnet01$ git lfs ls-files --size
f1753a034b * large-file-1.bin (106 MB)
2338a970c0 * large-file-2.bin (107 MB)
cb91d5908e * large-file-3.bin (108 MB)
hifzashakeel@hifza-ubuntu-server:~/Assignmnet01$
```

6.3 Commit and Push the LFS Files

Commit the files:

```
git commit -m "Add three files using Git LFS"
```

Push them to GitHub:

```
git push github main
```

Wait for the actual upload to finish successfully. The assignment asks you to capture the successful LFS upload, preferably showing `100% (3/3)`.

Screenshot 9: Git LFS Push

Filename: `task3_lfs_push.png`

```
hifzashakeel@hifza-ubuntu-server:~/Assignmnet01$ git commit -m "add three files using git lfs"
[main e14f005] add three files using git lfs
 4 files changed, 10 insertions(+)
 create mode 100644 .gitattributes
 create mode 100644 large-file-1.bin
 create mode 100644 large-file-2.bin
 create mode 100644 large-file-3.bin
hifzashakeel@hifza-ubuntu-server:~/Assignmnet01$ git push github main
```

Task 3 Result: After successful completion, three files larger than 100 MiB will be tracked using Git LFS and pushed to GitHub.

7. Task 4: Create a Portfolio or CV with GitHub Pages

7.1 Create the Portfolio Repository

Create a new public GitHub repository named:

```
hifzashakeel52.github.io
```

Create a personal portfolio using HTML and CSS. It must contain these files:

- `index.html`
- `styles.css`

The portfolio must include the following sections:

1. About Me
2. Education
3. Skills
4. Projects

Replace any sample text with accurate information about yourself before publishing.

If you create the website locally, initialize the repository and push it to GitHub:

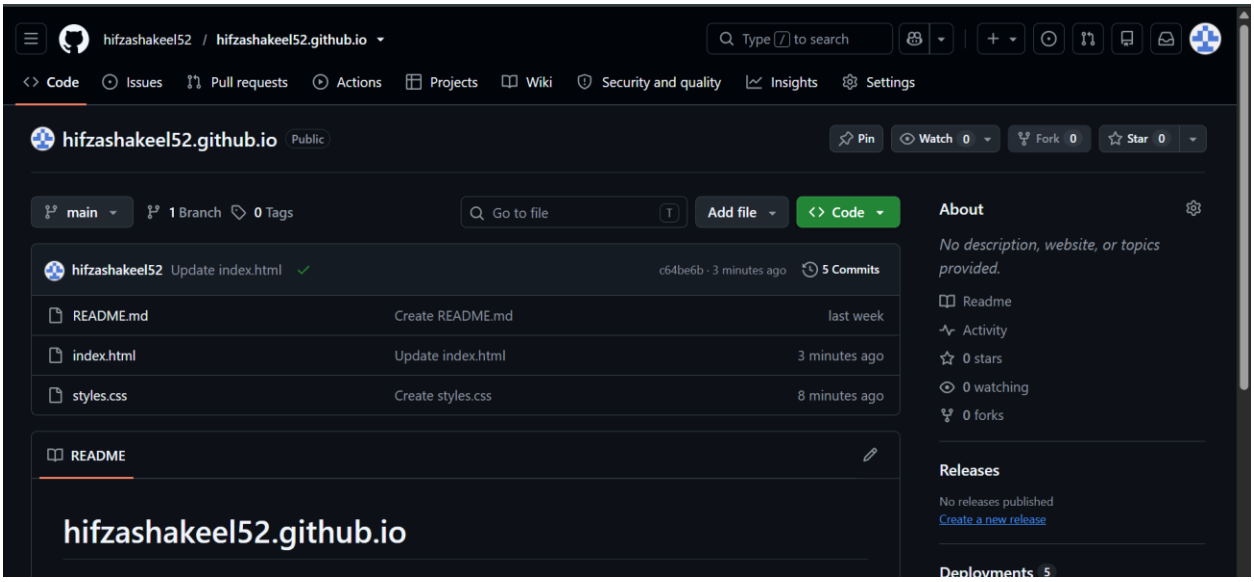
```
git init
git branch -M main
git add index.html styles.css
git commit -m "Create personal portfolio"
git remote add origin
https://github.com/hifzashakeel52/hifzashakeel52.github.io.git
git push -u origin main
```

Run these commands in the portfolio project directory, not inside the `Assignment01` repository. If the repository already exists locally or has already been initialized, do not repeat the initialization or remote-add commands unnecessarily.

Open the portfolio repository on GitHub and verify that `index.html` and `styles.css` are visible.

Screenshot 10: Portfolio Repository

Filename: `task4_pages_repository.png`



7.2 Publish the Portfolio with GitHub Pages

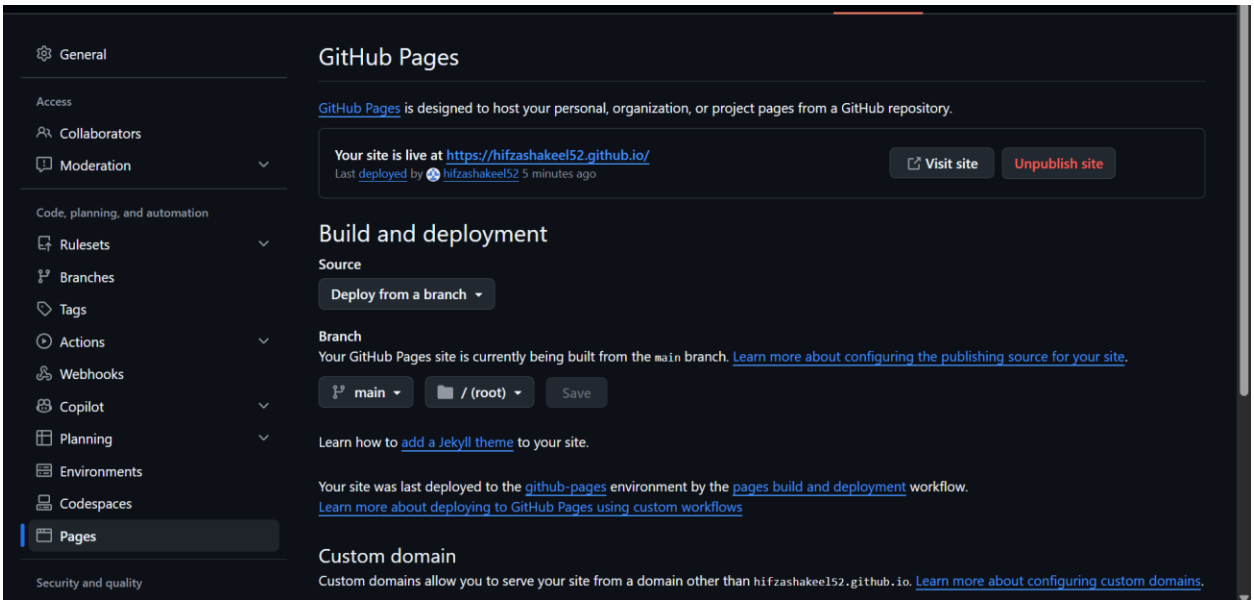
Open the portfolio repository on GitHub.

1. Select **Settings**.
2. Open **Pages**.
3. Under **Build and deployment**, select **Deploy from a branch**.
4. Select the `main` branch.
5. Select the `/ (root)` folder.
6. Click **Save**.
7. Wait until GitHub reports that the website is published or live.

The deployment screen should show the GitHub Pages status and website URL.

Screenshot 11: GitHub Pages Deployment

Filename: `task4_pages_deployment.png`



7.3 Verify the Live Portfolio

Open:

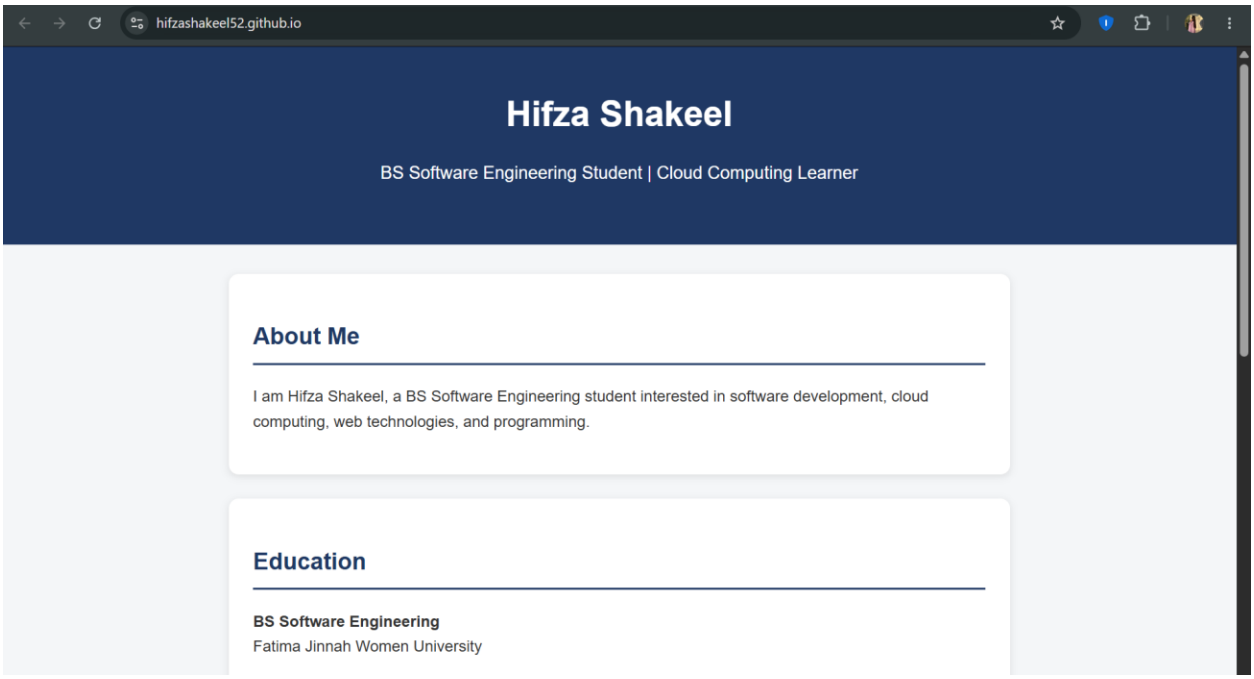
`https://hifzashakeel52.github.io/`

Confirm that:

- The website loads successfully.
- The full GitHub Pages URL is visible.
- At least one required section, such as About Me, Education, Skills, or Projects, is visible.

Screenshot 12: Live Portfolio

Filename:



Task 4 Result: After successful completion, the portfolio will be stored in a public GitHub repository and published through GitHub Pages.

8. Assignment Summary

This assignment demonstrates practical skills in Git, Gitea, GitHub, Git LFS, and GitHub Pages.

The completed work should demonstrate that:

- Gitea was installed and run on an Ubuntu server.
- A local repository was pushed to Gitea.
- The same repository was pushed to GitHub using a second remote.
- Three files larger than 100 MiB were tracked with Git LFS and pushed.
- A personal portfolio was created and published using GitHub Pages.
- The required assignment files and screenshots were prepared for submission.

Record only results that you have actually verified.

9. Final Submission Checklist

Before submitting, confirm the following:

- yes

  `Assignment01.md` is inside `CC/Assignments/Assignment01/`.

- [ ] yes

  `Assignment01_Solution.docx` opens correctly.

- [ ] yes

  `Assignment01_Solution.pdf` matches the Word document.

- [ ] yes

  The `screenshots/` directory contains all 12 readable screenshots.

- [ ] yes

  Each screenshot was captured from my own practical work.

- [ ] yes

  My GitHub username is visible in every Ubuntu terminal screenshot.

- [ ] yes

  No screenshot contains a password, token, or private key.

- [ ] yes

  My `CC` repository is public.

- [ ] yes

  My GitHub `Assignment01` repository is public.

- [ ] yes

  My `hifzashakeel52.github.io` repository is public.

- [ ] yes

  My Gitea repository was created and pushed from the Ubuntu server.

- [ ] yes

  All repositories use the `main` branch.

- yes

  The complete `Assignments/Assignment01` directory was pushed to `CC`.

- yes

  All three required assignment files were uploaded to Google Classroom.

- yes

  The three required links were submitted on Google Classroom.

End of Assignment 01